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Touch display panel

Abstract

A touch display panel is provided. The angle between an opening direction of a fracture and a grid line of a grid of a grid electrode in the present application is arranged to be acute an angle, so that the opening direction of the fracture is not perpendicular to the grid line of the grid of the grid electrode, and the opening of the fracture is narrowed, thereby reducing a light leakage of a plurality of sub-pixels located in the grid, which reduces a display panel color shift, and thus reducing a problem of uneven display caused by the display panel color shift.

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Background/Summary

FIELD OF DISCLOSURE
(1) The present disclosure relates to the field of display technology, and more particularly, to a touch display panel.
BACKGROUND OF DISCLOSURE
(2) A touch technology of a display panel has gradually developed from an add-on touch panel (TP) technology to a direct on thin film encapsulation touch (DOT) layer with a touch sensor on the liquid crystal panel. Compared with the add-on TP, DOT is directly integrated on the display panel, no separate flexible circuit board is required, a cost is lower, and further facilitates achieving flexible bending of the display panel.
(3) A conventional DOT technology is a metal mesh structure adopting metal mesh lines, i.e., achieving capturing and identifying touch signals through a mutual capacitance sensing of the metal mesh lines after exposure and etching, and having sub-pixels located in the metal mesh to prevent light of the sub-pixels from emitting. In a current display panel of a mutual capacitance mode, driving electrodes and sensing electrodes are obtained by cutting a portion of the mesh lines in an entire metal mesh, and dummy electrodes are also provided. However, in the portion of the mesh lines that are cut, a fracture region formed through cutting and other regions of the metal grid in addition to the fracture region have different reflection effects on the light emitted by the sub-pixels and an external ambient light. Specifically, the fracture region does not reflect light, and the other regions of the metal mesh in addition to the fracture region easily reflect light. That is to say, the light emitted by the sub-pixels located in the metal mesh with the fracture is different from light emitted by the sub-pixels located in the metal mesh without the fracture, causing a color shift and uneven display is likely to occur on a display screen.
SUMMARY
(4) The present application provides a touch display panel to solve a problem of uneven display caused by a display panel color shift.
(5) The present application provides a touch display panel including: a display light-emitting layer, wherein the display light-emitting layer includes a plurality of sub-pixels arranged at intervals; and a grid electrode, wherein the grid electrode is arranged on a light-emitting side of the display light-emitting layer, and the plurality of sub-pixels are located in a grid of the grid electrode; wherein, a portion of the grid of the grid electrode is provided with a fracture, the fracture includes a first section and a second section, an angle between the first section and a grid line of the portion of the grid with the fracture of the grid electrode and an angle between the second section and the grid line of the portion of the grid with the fracture of the grid electrode is are each an acute angle.
(6) Optionally, in some embodiments of the present application, an orthographic projection of the first section in a direction perpendicular to the grid line of the grid electrode and an orthographic projection of the second section in the direction perpendicular to the grid line of the grid electrode

are overlapped on an overlapping portion, and an orthographic projection of a center point of each of the plurality of sub-pixels in the direction perpendicular to the grid line of the grid electrode is located within a range of the overlapping portion.

(7) Optionally, in some embodiments of the present application, an extension plane of the first section and an extension plane of the second section in an opening direction of the fracture are neither intersected nor tangent to the plurality of sub-pixels on a plane.

(8) Optionally, in some embodiments of the present application, a first channel and a second channel are formed between the first section and the second section, wherein an angle is between the first channel and the second channel.

(9) Optionally, in some embodiments of the present application, a first concave portion is provided on the first section, and a first protrusion portion protruding toward the first concave portion is provided on the second section.

(10) Optionally, in some embodiments of the present application, a width of a cross-sectional shape of the first concave portion in a thickness direction of the display light-emitting layer is arranged ascendingly in a direction from the first section to the second section, and a width of a cross-sectional shape of the first protrusion in the thickness direction of the display light-emitting layer is arranged ascendingly in the direction from the first section to the second section.

(11) Optionally, in some embodiments of the present application, the cross-sectional shape of the first concave portion and the cross-sectional shape of the first protrusion portion in the thickness direction of the display light-emitting layer are square, and the width of the cross-sectional shape of the first concave portion in the thickness direction of the display light-emitting layer is greater than the width of the cross-sectional shape of the first protrusion portion in the thickness direction of the display light-emitting layer.

(12) Optionally, in some embodiments of the present application, an end of the first protruding portion is located in the first concave portion.

(13) Optionally, in some embodiments of the present application, a second protrusion portion protruding toward a direction of the second section is provided on the first section.

(14) Optionally, in some embodiments of the present application, an orthographic projection of the first protrusion portion in a direction parallel to the first section or the second section and an orthographic projection of the second protrusion portion in the direction parallel to the first section or the second section at least partially overlap.

(15) Correspondingly, the present application further provides a touch display panel including: a display light-emitting layer, wherein the display light-emitting layer includes a plurality of sub-pixels arranged at intervals; and a grid electrode, wherein the grid electrode is arranged on a light-emitting side of the display light-emitting layer, and the plurality of sub-pixels are located in a grid of the grid electrode; wherein, a portion of the grid of the grid electrode is provided with a fracture, the fracture includes a first section and a second section, and an angle between the first section and a grid line of the portion of the grid with the fracture of the grid electrode and an angle between the second section and the grid line of the portion of the grid with the fracture of the grid electrode are each an acute angle; and the angle between the first section and the grid line of the portion of the grid with the fracture of the grid electrode and the angle between the second section and the grid line of the portion of the grid with the fracture of the grid electrode is are both greater than or equal to 30 degrees and less than 90 degrees.

(16) Optionally, in some embodiments of the present application, an orthographic projection of the first section in a direction perpendicular to the grid line of the grid electrode and an orthographic projection of the second section in the direction perpendicular to the grid line of the grid electrode are overlapped on an overlapping portion, and an orthographic projection of a center point of each of the plurality of sub-pixels in the direction perpendicular to the grid line of the grid electrode is located within a range of the overlapping portion.

(17) Optionally, in some embodiments of the present application, an extension plane of the first

section and an extension plane of the second section in an opening direction of the fracture are neither intersected nor tangent to the plurality of sub-pixels on a plane.
(18) Optionally, in some embodiments of the present application, a first channel and a second channel are formed between the first section and the second section, wherein an angle is between the first channel and the second channel.
(19) Optionally, in some embodiments of the present application, a first concave portion is provided on the first section, and a first protrusion portion protruding toward the first concave portion is provided on the second section.
(20) Optionally, in some embodiments of the present application, a width of a cross-sectional shape of the first concave portion in a thickness direction of the display light-emitting layer is arranged ascendingly in a direction from the first section to the second section, and a width of a cross-sectional shape of the first protrusion in the thickness direction of the display light-emitting layer is arranged ascendingly in the direction from the first section to the second section.
(21) Optionally, in some embodiments of the present application, the cross-sectional shape of the first concave portion and the cross-sectional shape of the first protrusion portion in the thickness direction of the display light-emitting layer are square, and the width of the cross-sectional shape of the first concave portion in the thickness direction of the display light-emitting layer is greater than the width of the cross-sectional shape of the first protrusion portion in the thickness direction of the display light-emitting layer.
(22) Optionally, in some embodiments of the present application, an end of the first protruding portion is located in the first concave portion.
(23) Optionally, in some embodiments of the present application, a second protrusion portion protruding toward a direction of the second section is provided on the first section.
(24) Optionally, in some embodiments of the present application, an orthographic projection of the first protrusion portion in a direction parallel to the first section or the second section and an orthographic projection of the second protrusion portion in the direction parallel to the first section or the second section at least partially overlap.
(25) The application provides a touch display panel, wherein the touch display panel includes: a display light-emitting layer, wherein the display light-emitting layer includes a plurality of sub-pixels arranged at intervals; a grid electrode, wherein the grid electrode is arranged on a light-emitting side of the display light-emitting layer, and the plurality of sub-pixels are located in a grid of the grid electrode; wherein, a portion of the grid of the grid electrode is provided with a fracture, the fracture includes a first section and a second section, and an angle between the first section and the second section and a grid line of the portion of the grid with the fracture of the grid electrode and an angle between the second section and the grid line of the portion of the grid with the fracture of the grid electrode are each an acute angle. In the present application, the angle between an opening direction of the fracture and the grid line of the portion of the grid with the fracture of the grid electrode is arranged to be the acute angle, so that the opening direction of the fracture is not perpendicular to the grid line of the portion of the grid with the fracture of the grid electrode, and the opening of the fracture is narrowed, thereby reducing a light leakage of a plurality of sub-pixels located in the grid, which reduces a display panel color shift, and thus reducing a problem of uneven display caused by the display panel color shift.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) In order to describe technical solutions in the present invention clearly, drawings to be used in the description of embodiments will be described briefly below. Obviously, drawings described below are only for some embodiments of the present invention, and other drawings can be obtained

by those skilled in the art based on these drawings without creative efforts.
(2) FIG. **1** is a schematic view of a touch display panel provided by a first embodiment of the present application.
(3) FIG. **2** is a cross-sectional view of a line A-B in FIG. **1**.
(4) FIG. **3** is a partial enlarged view of FIG. **1**.
(5) FIG. **4** is a schematic view of a touch display panel provided by a second embodiment of the present application.
(6) FIG. **5** is a partial enlarged view of FIG. **4**.
(7) FIG. **6** is a schematic view of a first concave portion and a first protrusion portion of the touch display panel provided by the second embodiment of the present application.
(8) FIG. **7** is a schematic view of a touch display panel provided by a third embodiment of the present application.
(9) FIG. **8** is a schematic view of a touch display panel provided by a fourth embodiment of the present application.
(10) FIG. **9** is a schematic view of a touch display panel provided by a fifth embodiment of the present application.
(11) FIG. **10** is a schematic view of a touch display panel provided by a sixth embodiment of the present application.
(12) FIG. **11** is a schematic view of a touch display panel provided by a seventh embodiment of the present application.
DETAILED DESCRIPTION OF PRESENT EMBODIMENTS
(13) The technical solution of the present application embodiment will be clarified and completely described with reference accompanying drawings in embodiments of the present application embodiment. Obviously, the present application described parts of embodiments instead of all of the embodiments. Based on the embodiments of the present application, other embodiments which can be obtained by a skilled in the art without creative efforts fall into the protected scope of the present application.
(14) In the description of the present application, it should be explained that the terms “center”, “portrait”, “transverse”, “length”, “width”, “thickness”, “upper”, “lower”, “front”, the directions or positional relationships indicated by “back”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inside”, “outside”, etc. are based on the drawings. The orientation or positional relationship is only for the convenience of describing the present application and simplifying the description, and does not indicate or imply that the device or element referred to must have a specific orientation, or a structure or an operation in a specific orientation, and should not be viewed as limitations of the present application. In addition, terms “first” and “second” are used for descriptive purposes only, and cannot be understood as indicating or implying relative importance or implicitly indicating the number of technical features indicated. Therefore, the features defined as “first” and “second” may explicitly or implicitly include one or more of the features. In the description of the present application, the meaning of “multiple” is two or more, unless specifically defined otherwise.
(15) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. The following descriptions are presented to enable any person skilled in the art to make and use the application. Descriptions of specific embodiments and applications are provided only as examples. Various modifications and combinations of the examples described herein will be readily apparent to those skilled in the art, and the general principles defined herein may be applied to other examples and applications without departing from the spirit and scope of the application. Thus, the present application is not intended to be limited to the examples described and shown, but is to be accorded the widest scope consistent with the principles and features disclosed herein. Unless otherwise

specified, parallelism or perpendicularity in orientations involved in the present application are not parallelism or perpendicularity in a strict sense, as long as a corresponding structure can achieve a corresponding purpose.
(16) The present application provides a touch display panel, which will be described in detail below. It should be noted that a description order of the following embodiments is not intended to limit a preferred order of embodiments of the present application. The touch display panel of the present application can be utilized in both liquid crystal display devices and OLED display devices.
(17) Referring to FIGS. **1-3**, FIG. **1** is a schematic view of a touch display panel **100** provided by a first embodiment of the present application, FIG. **2** is a cross-sectional view of a line A-B in FIG. **1**, and FIG. **3** is a partial enlarged view of FIG. **1**. An embodiment of the present application provides the touch display panel **100**, which includes: an array substrate, a display light-emitting layer **10**, and a grid electrode **20**.
(18) The display light-emitting layer **10** is disposed on the array substrate, and the display light-emitting layer **10** includes a plurality of sub-pixels **11** arranged at intervals. The grid electrode **20** is disposed on a light-emitting side of the display light-emitting layer **10**, and the plurality of sub-pixels **11** are located in a grid **21** of the grid electrode **20**.
(19) The array substrate is a thin-film transistor array substrate commonly used in the prior art, and when arranging, the array substrate can be arranged according to the prior art, which will not be described in detail herein. In an embodiment of the present application, organic light-emitting display is used as a specific implementation manner for description. Specifically, the display light-emitting layer **10** includes an organic light-emitting layer, and the organic light-emitting layer includes the plurality of sub-pixels **11** arranged in an array, and the grid electrode **20** is disposed on a light-emitting side of the organic light-emitting layer. In order to achieve a touch function, electrodes are further arranged on the display light-emitting layer **10**, i.e., the grid electrode **20** in the embodiments of the present application.
(20) Specifically, a plurality of grids **21** of the grid electrode **20** are in one-to-one correspondences with the plurality of sub-pixels **11**. That is say, a sub-pixel **11** is provided in each of the plurality of grids **21** of the grid electrode **20**. The light-emitting layer **10** includes a plurality of pixel units, the plurality of pixel units are arranged in an array to form the display light-emitting layer **10** of the display panel, and the plurality of pixel units include the plurality of light-emitting sub-pixels **11**. Specifically, the plurality of light-emitting sub-pixels **11** can include a green sub-pixel **111**, a red sub-pixel **112**, and a blue sub-pixel **113**. A region formed by each of the plurality of light-emitting sub-pixels **11** is a light-emitting region, and a region formed between two adjacent sub-pixels **11** is a non-light-emitting region. For example, a region formed between the green sub-pixel **111** and the red sub-pixel **112** is a non-light-emitting region. The grid electrode **20** is arranged in non-light-emitting regions formed between each of the plurality of light-emitting sub-pixels **11**. After the grid electrode **20** is arranged, the plurality of sub-pixels **11** are located in the plurality of grids **21** of the grid electrode **20**. In addition, an array arrangement of the plurality of sub-pixels **11** can be any one of a pinwheel arrangement, a tripod arrangement, a pearl arrangement, and a diamond arrangement, and can also be any other types of array arrangement.
(21) A portion of the plurality of grids **21** of the grid electrode **20** are each provided with a fracture **22**, and an angle between an opening direction of the fracture **22** and a grid line **25** of the portion of the plurality of grids **21** with the fracture **22** of the grid electrode **20** is an acute angle. Specifically, the fracture **22** includes a first section **23** and a second section **24**, and the angle P between the first section **23** and the grid line **25** of the portion of the plurality of grids **21** with the fracture **22** of the grid electrode **20** and the angle P between the second section **24** and the grid line **25** of the portion of the plurality of grids **21** with the fracture **22** of the grid electrode **20** are each an acute angle.
(22) That is to say, the present application sets the angle between the opening direction of the fracture **22** and the grid line **25** of the portion of the plurality of grids **21** with the fracture **22** of the grid electrode **20** as an acute angle, so that the opening direction of the fracture **22** is not

perpendicular to the grid line **25** of a grid **21** with the fracture **22**, and can therefore narrow an opening of the fracture **22**, thereby reducing light leakage of the plurality of sub-pixels **111** located in the plurality of grids **21**, which can reduce a display panel color shift, and thus reducing a problem of uneven display caused by the display panel color shift.

(23) Moreover, the angle between the opening direction of the fracture **22** and the grid line **25** of the grid **21** with the fracture **22** of the grid electrode **20** is greater than or equal to 30 degrees and less than 90 degrees. Specifically, the angle P between the opening direction of the fracture **22** and the grid line **25** of the grid **21** with the fracture **22** of the grid electrode **20** is 34 degrees.

(24) Furthermore, in some embodiments, an orthographic projection of the first section **23** in a direction perpendicular to the grid line **25** of the grid electrode **20** and an orthographic projection of the second section **24** in the direction perpendicular to the grid line **25** of the grid electrode **20** are overlapped on an overlapping portion **26**, and an orthographic projection of a center point of the sub-pixel **11** in the direction perpendicular to the grid line **25** of the grid electrode **20** is located within a range of the overlapping portion **26**. The orthographic projection of the center point of the sub-pixel **11** in the direction perpendicular to the grid line **25** of the grid electrode **20** is arranged within the range of the overlapping portion **26**, so that most light of the plurality of sub-pixels **11** located in the plurality of grids **21** of the grid electrode **20** cannot be leaked through the fracture **22**, which can further reduce the display panel color shift, and thus further reduce the problem of the uneven display caused by the display panel color shift.

(25) Furthermore, in some embodiments, an extension plane of the first section **23** and an extension plane of the second section **24** in the opening direction of the fracture **22** are neither intersected nor tangent to the sub-pixel **11** on a plane. Since the extension plane of the first section **23** and the extension plane of the second section **24** in the opening direction of the fracture **22** are neither intersected nor tangent to the sub-pixel **11** on the plane, the opening direction of the fracture **22** can be avoided from the plurality of sub-pixels **11**, so that the light of the plurality of sub-pixels **11** located in the grid **21** is not emitted from the fracture **22**, which can reduce the display panel color shift, and thus solve the problem of the uneven display caused by the display panel color shift.

(26) Referring to FIGS. **4-5**, FIG. **4** is a schematic view of a touch display panel **100** provided by a second embodiment of the present application, and FIG. **5** is a partial enlarged view of FIG. **4**. A difference between this embodiment and the touch display panel **100** provided in FIG. **1** is that a first channel **221** and a second channel **222** are formed between the first section **23** and the second section **24**, and an angle is between the first channel **221** and the second channel **222**.

(27) Furthermore, the first section **23** is provided with a first concave portion **231**, and the second section **24** is provided with a first protrusion portion **241** protruding toward the first concave portion **231**.

(28) In the present application, electrodes are formed on the grid electrode **20** by providing the fracture **22** in a portion of the plurality of grids **21** of the grid electrode **20**. Since adjacent electrodes is required to be insulated, the fracture **22** is required to have a certain degree of opening. In addition, the greater the opening of the fracture **22**, the better an insulation between the adjacent electrodes. However, the fracture **22** causes the light of the plurality of sub-pixels **11** located in the plurality of grids **21** with the fracture **22** to leak out, which causes the display panel color shift, resulting in the uneven display of a display screen.

(29) In the present application, the first concave portion **231** and the first protrusion portion **241** are respectively provided on the first section **23** and the second section **24** of the fracture **22**, and the first channel **221** and the second channel **222** are formed on the fracture **22** between the first concave portion **231** and the first protrusion portion **241**. An angle is between the first channel **221** and the second channel **222**, and the angle is greater than 0 degrees and less than 180 degrees, preferably, the angle ranges from 45 degrees to 180 degrees. Therefore, the angle between the opening direction of the fracture **22** and the grid line **25** of the grid **21** with the fracture **22** of the grid electrode **20** can be an acute angle, and the opening direction of the fracture **22** is not

perpendicular to the grid line **25** of the grid **21** with the fracture **22** of the grid electrode **20**.
(30) The light of the plurality of sub-pixels **11** is blocked by the first concave part **231** and the first protrusion portion **241** in directions except for a direction parallel to the first section **23** or the second section **24**. Since the angle is between the first channel **221** and the second channel **222**, the light of the plurality of sub-pixels **11** entering the first channel **221** from the second channel **222** and is blocked by the first concave portion **231**, thereby greatly reducing the light leakage of the plurality of sub-pixels **11** of the touch display panel **100** at the fracture **22**. When the orthographic projection of the first concave portion **231** and the orthographic projection of the first protrusion portion **241** in the direction parallel to the first section **23** or the second section **24** overlap, the present application can prevent the light of the plurality of sub-pixels **11** of the touch display panel **100** from leaking at the fracture **22**.
(31) In addition, the first concave portion **231** and the first protrusion portion **241** are utilized to keep an opening and an insulation of the fracture **22**, so the first concave portion **231** and the first protrusion portion **241** can prevent the light of the plurality of sub-pixels **11** located in the grid **21** with the fracture **22** from leaking, so as to reduce the display panel color shift, and thus reducing the problem of the uneven display caused by the display panel color shift.
(32) In some embodiments, a length direction of the first concave portion **231** and a length direction of the first protrusion portion **241** are parallel to a thickness direction of the display light-emitting layer **10**. A length of the first concave portion **231** and a length of the first protrusion portion **241** are arranged along the thickness direction of the display light-emitting layer **10** to have a better light blocking effect, rather than being arranged perpendicular to the thickness direction the display light-emitting layer **10**.
(33) In addition, a width of a cross-sectional shape of the first concave portion **231** in the thickness direction of the display light-emitting layer **10** is arranged ascendingly in a direction from the first section **23** to the second section **24**. A width of a cross-sectional shape of a protrusion portion **241** in the thickness direction of the display light-emitting layer **10** is arranged ascendingly in the direction from the first section **23** to the second section **24**.
(34) Specifically, referring to FIG. **6**, which is a schematic view of the first concave portion and the first protrusion portion of the touch display panel **100** provided by the second embodiment of the present application. FIG. **6** provides three types of structures of the first concave portion and the first protrusion portion (A), (B) and (C). Referring to type (A) of FIG. **6**, the cross-sectional shape of the first concave portion **231** and the cross-sectional shape of the first protrusion portion **241** in the thickness direction of the display light-emitting layer **10** is a triangle. Referring to type (B) of FIG. **6**, the cross-sectional shape of the first concave portion **231** and the cross-sectional shape of the first protrusion portion **241** in the thickness direction of the display light-emitting layer **10** is a semicircle. Referring to type (C) of FIG. **6**, the cross-sectional shape of the first concave portion **231** and the cross-sectional shape of the first protrusion portion **241** in the thickness direction of the display light-emitting layer **10** is a trapezoid.
(35) That is to say, through the above-mentioned arrangements, a small end of the first protrusion portion **241** protrudes toward a direction of the first concave portion **231** in the present application. Therefore, since a width of the small end of the first protrusion portion **241** is smaller and a width of a large end of the first concave portion **231** is wider, a larger distance can be between the first protrusion portion **241** and the first concave portion **231**, so that the fracture **22** between adjacent electrodes has a larger opening and a greater insulation. In this way, the small end of the first protrusion portion **241** can also be arranged to be closer to the first section **23**, so that the first protrusion portion **241** has a better light blocking effect.
(36) In some embodiments, the first concave portion **231** penetrates through two sides of the first section **23** in the thickness direction of the display light-emitting layer **10**. Correspondingly, the two sides of the first protrusion portion **241** in the thickness direction of the display light-emitting layer **10** are respectively flush with two sides of the first section **23** or two sides of the second section **24**

in the thickness direction of the display light-emitting layer **10**. Therefore, the first protrusion **241** can block the light of the plurality of sub-pixels **11** emitting from the fracture **22** with a maximum area in the thickness direction of the display light-emitting layer **10**.
(37) Referring to FIG. **7**, which is a schematic view of a touch display panel **100** provided by a third embodiment of the present application. In this embodiment, the cross-sectional shape of the first concave portion **231** and the cross-sectional shape of the first protrusion portion **241** in the thickness direction of the display light-emitting layer **10** are square, and a width of the cross-sectional shape of the first concave portion **231** is greater than a width of the cross-sectional shape of the first protrusion portion **241** in the thickness direction of the display light-emitting layer **10**. That is to say, in this embodiment, the width of the cross-sectional shape of the first concave portion **231** and the width of the first protrusion portion **241** in the thickness direction of the display light-emitting layer **10** do not change in the direction from the first section **23** to the second section **24**. However, when the width of the cross-sectional shape of the first concave portion **231** in the thickness direction of the display light-emitting layer **10** is larger than the width of the cross-sectional shape of the first protrusion portion **241** in the thickness direction of the display light-emitting layer **10** can also allow the fracture **22** between the adjacent electrodes to have the larger opening and the greater insulation.
(38) Furthermore, referring to FIG. **8**, which is a schematic view of a touch display panel **100** provided by a fourth embodiment of the present application. A difference between this embodiment and the touch display panel **100** provided in FIG. **4** is that an end of the first protrusion portion **241** is located in the first concave portion **231**. An end of the first protrusion portion **241** is extended into the first concave portion **231**, and the first protrusion portion **241** covers a cross-sectional surface of the fracture **22** in an exit direction F of the light of the sub-pixel **11** from the fracture **22**, so that the first protrusion **241** can block the light of the plurality of sub-pixels **11** emitting from the fracture **22** with the maximum area.
(39) Furthermore, referring to FIG. **9**, which is a schematic view of a touch display panel **100** provided by a fifth embodiment of the present application. A difference between this embodiment and the touch display panel **100** provided in FIG. **8** is that the second section **24** is provided with a second protrusion portion **242** protruding in the direction of the first section **23**. The first protrusion portion **241** and the second protrusion portion **242** are arranged at intervals along a direction F parallel to the first section **23** or the second section **24**.
(40) A length of the first protrusion portion **241** in a protruding direction is greater than a length of the second protrusion portion **242** in a protruding direction.
(41) On a basis of an end of the first protrusion portion **241** extending into the first concave portion **231**, the second section **24** is provided with a second protrusion protruding in the direction of the first section **23**. The length of the first protrusion portion **241** in the protruding direction is greater than the length of the second protrusion portion **242** in the protruding direction. Although the first protrusion portion **241** can block the light of the plurality of sub-pixels **11** emitting from the fracture **22** with the maximum area, a portion of the light of the plurality of sub-pixels **11** is still emitted from a gap between the first protrusion portion **241** and the first concave portion **231**. Therefore, the second protrusion portion **242** is provided for reducing the light of the plurality of sub-pixels **11** from entering the gap between the first protrusion portion **241** and the first concave portion **231**, which can reduce an amount of light entering the gap between the first protrusion portion **241** and the first concave portion **231**, thereby further reducing the light leakage of the plurality of sub-pixels **11**.
(42) Furthermore, the first section **23** is provided with a second concave portion **232** at a position in an extending direction of the second protrusion portion **242**. Through having the second concave portion **232**, a distance between the first section **23** and the second protrusion portion **242** can be increased, thereby further increasing an insulation between the first section **23** and the second protrusion portion **242**.

(43) In addition, a width of the cross-sectional shape of the second concave portion **232** in the thickness direction of the display light-emitting layer **10** is arranged ascendingly in the direction from the first section **23** to the second section **24**. A width of a cross-sectional shape of the second protrusion portion **242** in the thickness direction of the display light-emitting layer **10** is arranged ascendingly in the direction from the first section **23** to the second section **24**.
(44) The cross-sectional shape of the second concave portion **232** and the cross-sectional shape of the second protrusion portion **242** in the thickness direction of the display light-emitting layer **10** are triangles.
(45) That is to say, through the above-mentioned arrangements, a small end of the second protrusion portion **242** protrudes toward a direction of the second concave portion **232** in the present application. Therefore, since a width of the small end of the second protrusion portion **242** is smaller and a width of a large end of the second concave portion **232** is wider, a larger distance can be between the second protrusion portion **242** and the second concave portion **232**, so that the fracture **22** between the adjacent electrodes has the larger opening and the greater insulation. In this way, the small end of the second protrusion portion **242** can also be arranged to be closer to the first section **23**, so that the second protrusion portion **242** has a better light blocking effect.
(46) Referring to FIG. **10**, which is a schematic view of a touch display panel **100** provided by a sixth embodiment of the present application. A difference between the present embodiment and the touch display panel **100** provided in FIG. **4** is that the first section **23** is provided with a second protrusion portion **232** protruding toward the second section **24**.
(47) That is to say, in the present application, the second protrusion portion **232** is provided on the first section **23**. Since the first protrusion portion **241** and the second protrusion portion **232** respectively protrude in opposite directions, the first protrusion portion **241** and the second protrusion portion together form a blocking wall at the fracture **22** to block the light of the plurality of sub-pixels **11** emitting from the fracture **22** from leaking, so as to further reduce the light leakage of the plurality of sub-pixels **11** without affecting the insulation between the first section **23** and the second section **24**.
(48) Furthermore, in some embodiments, an orthographic projection of the first protrusion portion **241** in the direction F parallel to the first section **23** or the second section **24** and an orthographic projection of the second protrusion portion **232** in the direction F parallel to the first section **23** or the second section **24** at least partially overlap.
(49) In the present application, the orthographic projection of the first protrusion portion **241** and the orthographic projection of the second protrusion portion **232** in a direction of the light of the plurality of sub-pixels **11** emitting from the fracture **22** are overlapped, and it is not required to have the first protrusion portion **241** arranged to be adjacent to the first section **23**, or have the second protrusion portion **232** arranged adjacent to the second section **24**. Therefore, a larger distance can be between the first protrusion portion **241** and the first section **23**, a larger distance can be between the second protrusion portion **232** and the second section **24**, and a better insulation can be between the first section **23** and the second section **24**.
(50) Furthermore, the second section **24** is provided with a second concave portion **242** at a position in an extending direction of the second protrusion portion **232**. In the present application, a distance between the second section **24** and the second protrusion portion **232** can be increased through arranging the second concave portion **242**, thereby further increasing the insulation between the second section **24** and the second protrusion portion **232**.
(51) Moreover, the first protrusion portion **241** and the second protrusion portion **232** are arranged in parallel, which can minimize a space occupied by the first protrusion portion **241** and the second protrusion portion **232**, and a larger distance can also be between the first protrusion portion **241** and the second protrusion portion **232**. In addition, the length of the first protrusion portion **241** in the protruding direction is greater than a length of the second protrusion portion **232** in the protruding direction.

(52) Specifically, having the length of the first protrusion portion **241** and the length of the second protrusion portion **232** arranged to be unequal can ensure that the orthographic projection of the first protrusion portion **241** and the orthographic projection of the second protrusion portion **232** in the direction parallel to the first section **23** or the second section **24** has an overlapping length, and, can prevent a distance between the second protrusion portion **232** and the second section **24** from being too small.
(53) Furthermore, an end of the first protrusion portion **241** is located in the first concave portion **231**, and an end of the second protrusion portion **232** is located outside the second concave portion **242**. The first protrusion portion **241** and the second protrusion portion **232** can be arranged at intervals along the direction parallel to the first section **23** or the second section **24**, or the second protrusion portion **232** and the second protrusion portion **241** are arranged at intervals along the direction parallel to the first section **23** or the second section **24**. In this embodiment, for convenience of description, the second protrusion portion **232** and the first protrusion portion **241** are arranged at intervals along the direction parallel to the first section **23** or the second section **24**, and based on this, a principle is explained that when the light of the plurality of sub-pixels **11** is emitted from the fracture **22**, the emitted light first bypasses the second protrusion portion **232**, then enters the gap between the first protrusion portion **241** and the first concave portion **231** along an overlapping portion of the first protrusion portion **241** and the second protrusion portion **232**, and then emitted from gap between the first concave portion **241** and the first concave portion **231** bypassing the first protrusion portion **241**, so that the light of the plurality of sub-pixels **11** are reflected multiple times during a process of being emitted from the fracture **22**, thereby further reducing the light leakage of the plurality of sub-pixels **11** from the fracture **22**.
(54) In addition, a width of a cross-sectional shape of the second concave portion **242** in the thickness direction of the display light-emitting layer **10** is arranged descendingly in the direction from the first section **23** to the second section **24**. a width of a cross-sectional shape of the second protrusion portion **232** in the thickness direction of the display light-emitting layer **10** is arranged descendingly in the direction from the first section **23** to the second section **24**.
(55) The cross-sectional shape of the second concave portion **242** and the cross-sectional shape of the second protrusion portion **232** in the thickness direction of the display light-emitting layer **10** are triangles.
(56) That is to say, through the above-mentioned arrangements, a small end of the second protrusion portion **232** protrudes toward a direction of the second concave portion **242** in the present application. Therefore, since a width of the small end of the second protrusion portion **232** is smaller and a width of a large end of the second concave portion **242** is wider, a larger distance can be between the second protrusion portion **232** and the second concave portion **242**, so that the fracture **22** between the adjacent electrodes has the larger opening and the greater insulation. In this way, the small end of the second protrusion portion **232** can also be arranged to be closer to the first section **23**, so that the second protrusion portion **232** has a better light blocking effect.
(57) Referring to FIG. **11**, which is a schematic view of a touch display panel **100** provided by a seventh embodiment of the present application. A difference between this embodiment and the touch display panel **100** provided in FIG. **10** is that the first section **23** includes a first connecting plane **233**, and the first connecting plane **233** is connected between the first concave portion **231** and the second protrusion portion **232**. The second section **24** includes a second connecting plane **243** arranged corresponding to the first connecting plane **233**, and the second connecting plane **243** is connected between the second concave portion **242** and the first protrusion portion **241**. Through having the first connecting plane **233** and the second connecting plane **243** arranged, a distance between the second protrusion portion **232** and the first protrusion portion **241** can be increased, thereby increasing an insulation between the second protrusion portion **232** and the first protrusion portion **241**.
(58) The present application further provides a touch display device including the above-mentioned

touch display panel **100**.

(59) A problem-solving principle of the touch display device is similar to a problem-solving principle of the above-mentioned touch display panel **100**. Therefore, implementations and beneficial effects of the touch display device can be referred to descriptions of the above-mentioned touch display panel **100**, and repeated details are not reiterated herein.

(60) The touch display panel provided by the present application is described in detail above, the specific examples of this document are used to explain principles and embodiments of the present application, and the description of embodiments above is only for helping to understand the present application. Meanwhile, those skilled in the art will be able to change the specific embodiments and the scope of the present application according to the idea of the present application. In the above, the content of the specification should not be construed as limiting the present application. Above all, the content of the specification should not be the limitation of the present application.

Claims

1. A touch display panel comprising: a display light-emitting layer, wherein the display light-emitting layer comprises a plurality of sub-pixels arranged at intervals; and a grid electrode, wherein the grid electrode is arranged on a light-emitting side of the display light-emitting layer, and the plurality of sub-pixels are located in a grid of the grid electrode; wherein, a portion of the grid of the grid electrode is provided with a fracture, the fracture comprises a first section and a second section, an angle between the first section and a grid line of the portion of the grid with the fracture of the grid electrode and an angle between the second section and the grid line of the portion of the grid with the fracture of the grid electrode are each an acute angle.
2. The touch display panel according to claim 1, wherein an orthographic projection of the first section in a direction perpendicular to the grid line of the grid electrode and an orthographic projection of the second section in the direction perpendicular to the grid line of the grid electrode are overlapped on an overlapping portion, and an orthographic projection of a center point of each of the plurality of sub-pixels in the direction perpendicular to the grid line of the grid electrode is located within a range of the overlapping portion.
3. The touch display panel according to claim 1, wherein an extension plane of the first section and an extension plane of the second section in an opening direction of the fracture are neither intersected nor tangent to the plurality of sub-pixels on a plane.
4. The touch display panel according to claim 1, wherein a first channel and a second channel are formed between the first section and the second section, wherein an angle is between the first channel and the second channel.
5. The touch display panel according to claim 1, wherein a first concave portion is provided on the first section, and a first protrusion portion protruding toward the first concave portion is provided on the second section.
6. The touch display panel according to claim 5, wherein a width of a cross-sectional shape of the first concave portion in a thickness direction of the display light-emitting layer is arranged ascendingly in a direction from the first section to the second section, and a width of a cross-sectional shape of the first protrusion in the thickness direction of the display light-emitting layer is arranged ascendingly in the direction from the first section to the second section.
7. The touch display panel according to claim 5, wherein the cross-sectional shape of the first concave portion and the cross-sectional shape of the first protrusion portion in the thickness direction of the display light-emitting layer are square, and the width of the cross-sectional shape of the first concave portion in the thickness direction of the display light-emitting layer is greater than the width of the cross-sectional shape of the first protrusion portion in the thickness direction of the display light-emitting layer.
8. The touch display panel according to claim 5, wherein an end of the first protruding portion is

located in the first concave portion.
9. The touch display panel according to claim 5, wherein a second protrusion portion protruding toward a direction of the second section is provided on the first section.
10. The touch display panel according to claim 9, wherein an orthographic projection of the first protrusion portion in a direction parallel to the first section or the second section and an orthographic projection of the second protrusion portion in the direction parallel to the first section or the second section at least partially overlap.
11. A touch display panel comprising: a display light-emitting layer, wherein the display light-emitting layer comprises a plurality of sub-pixels arranged at intervals; and a grid electrode, wherein the grid electrode is arranged on a light-emitting side of the display light-emitting layer, and the plurality of sub-pixels are located in a grid of the grid electrode; wherein, a portion of the grid of the grid electrode is provided with a fracture, the fracture comprises a first section and a second section, an angle between the first section and a grid line of the portion of the grid with the fracture of the grid electrode and an angle between the second section and the grid line of the portion of the grid with the fracture of the grid electrode are each an acute angle; and the angle between the first section and the grid line of the portion of the grid with the fracture of the grid electrode and the angle between the second section and the grid line of the portion of the grid with the fracture of the grid electrode is are both greater than or equal to 30 degrees and less than 90 degrees.
12. The touch display panel according to claim 11, wherein an orthographic projection of the first section in a direction perpendicular to the grid line of the grid electrode and an orthographic projection of the second section in the direction perpendicular to the grid line of the grid electrode are overlapped on an overlapping portion, and an orthographic projection of a center point of each of the plurality of sub-pixels in the direction perpendicular to the grid line of the grid electrode is located within a range of the overlapping portion.
13. The touch display panel according to claim 11, wherein an extension plane of the first section and an extension plane of the second section in an opening direction of the fracture are neither intersected nor tangent to the plurality of sub-pixels on a plane.
14. The touch display panel according to claim 11, wherein a first channel and a second channel are formed between the first section and the second section, wherein an angle is between the first channel and the second channel.
15. The touch display panel according to claim 11, wherein a first concave portion is provided on the first section, and a first protrusion portion protruding toward the first concave portion is provided on the second section.
16. The touch display panel according to claim 15, wherein a width of a cross-sectional shape of the first concave portion in a thickness direction of the display light-emitting layer is arranged ascendingly in a direction from the first section to the second section, and a width of a cross-sectional shape of the first protrusion in the thickness direction of the display light-emitting layer is arranged ascendingly in the direction from the first section to the second section.
17. The touch display panel according to claim 15, wherein the cross-sectional shape of the first concave portion and the cross-sectional shape of the first protrusion portion in the thickness direction of the display light-emitting layer are square, and the width of the cross-sectional shape of the first concave portion in the thickness direction of the display light-emitting layer is greater than the width of the cross-sectional shape of the first protrusion portion in the thickness direction of the display light-emitting layer.
18. The touch display panel according to claim 15, wherein an end of the first protruding portion is located in the first concave portion.
19. The touch display panel according to claim 15, wherein a second protrusion portion protruding toward a direction of the second section is provided on the first section.
20. The touch display panel according to claim 19, wherein an orthographic projection of the first

protrusion portion in a direction parallel to the first section or the second section and an orthographic projection of the second protrusion portion in the direction parallel to the first section or the second section at least partially overlap.